Recorded clean-clone and release-candidate smoke tests

Repository source: CLEAN_CLONE_CHECK.md

Record type: Reproducibility evidence retained with release v0.1.0
Release date: September 6, 2026
Canonical repository: https://github.com/swihart/minvol-degree3-spectral-bound

This document records two fresh-checkout tests. The first established a fully byte-clean baseline. The second tested the v0.1.0 release candidate and passed all mathematical, build, metadata, and PDF-validity checks; only two secondary rendered documentation PDFs differed by two bytes from their committed copies. Exact binary identity of generated PDFs is not a release requirement.

Results at a glance

Date	Commit	Result
2026-09-04	`96cea5727507`	All checks and builds passed; tracked tree remained byte-clean
2026-09-06	`ff5118cc05ad`	All substantive checks and builds passed; two rendered documentation PDFs differed by two bytes

Both runs used a fresh temporary clone and an isolated Python environment. The full Poppler PDF preflight was unavailable on the local Mac and was supplied by the corresponding green GitHub Actions document job.

Baseline byte-clean run

On September 4, commit `96cea57275070f21babf30477ab1b128ec0f5eb6` was cloned into a temporary directory. The isolated clone installed the pinned Python dependencies, ran all Python and R checks, rebuilt the paper and every Markdown-derived PDF, ran the metadata and checksum checks, and remained byte-clean. The recorded result was:

```
CLEAN CLONE CHECK: PASS WITH PDF PREFLIGHT DELEGATED TO GITHUB ACTIONS
Commit: 96cea57275070f21babf30477ab1b128ec0f5eb6
Repository: swihart/minvol-degree3-spectral-bound
PDF validation: basic local PDF and checksum checks passed; full PDF preflight delegated to GitHub
    Actions
Tracked working tree after all checks and rebuilds: clean
```

Release-candidate smoke test

On September 6, release-candidate commit `ff5118cc05ada29d26631c82d0cbf69613ed88c1` passed Markdown-source discovery, the PDF-canonicalizer regression, release preflights before and after building, all exact Python checks, both independent R checks, both document builds, the metadata preflight, basic validity checks for all 14 PDFs, and verification of the checksum manifest generated by that run.

The strict binary comparison found changes only in:

```
rendered/markdown/README.pdf                60287 -> 60285 bytes
rendered/markdown/proof/BRIDGE_LEMMAS.pdf   80259 -> 80261 bytes
rendered/markdown/SHA256SUMS.txt             corresponding hash updates
```

No Python, R, LaTeX, Markdown, release metadata, proof source, or mathematical output changed. The two PDF differences are consistent with renderer-level binary variability and are documented rather than treated as a mathematical or release-blocking failure.

Recorded local environment for the release-candidate run

Component	Recorded value
Operating system	macOS 26.2, Darwin 25.2.0, `x86_64`
Git	2.50.1 (Apple Git-155)
Python	3.13.15
NumPy	2.3.5
SymPy	1.14.0
R	4.6.0
Pandoc	2.19.2
XeTeX	3.141592653-2.6-0.999994, TeX Live 2022
`latexmk`	4.77
Poppler tools	Not installed locally; full preflight delegated to GitHub Actions

Machine-specific temporary paths and local artifact paths are omitted. The original run logs were retained locally under the Git-ignored `ci-artifacts/` directory.

Interpretation and limits

Together, these tests show that the mathematical programs run from a fresh clone; the exact Python and independent R checks pass; the manuscript and documentation sources compile; release and authorship metadata pass their checks; the generated PDFs are structurally valid and pass the hosted Poppler preflight; and no uncommitted source file is required.

The SHA-256 manifest authenticates the rendered PDF artifacts supplied with a particular release. It does not assert that every TeX/Pandoc invocation on every platform must produce identical PDF bytes.

These tests do **not** constitute independent subject-matter review, peer review, or certification that the proposed theorem is correct. The release remains an **AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification**. See `REPRODUCIBILITY.md` and `VERIFICATION_STATUS.md` for details.